



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: X
SUBJECT: BIOLOGY

FULL MARKS: 40
DATE: 14.7.22

General Instructions:

- The paper consists of three printed pages.
- Answers should be to the point.
- Copy the question number carefully before answering the questions.
- Marks will be deducted for spelling errors.

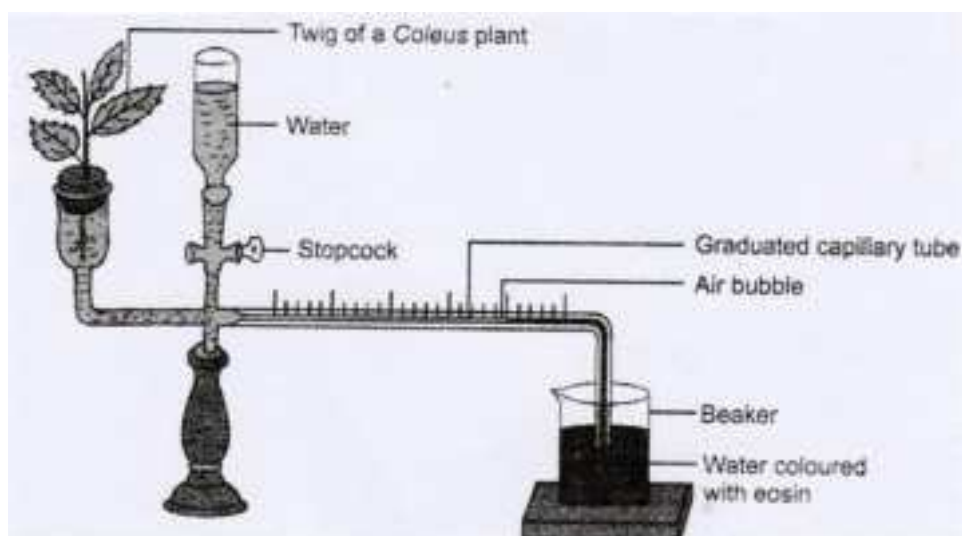
1. Choose the correct answer: [5]

- a. The space between the cell wall and plasma membrane in a plasmolysed cell is filled with
i. Isotonic solution, ii. Hypotonic solution, iii. Hypertonic solution,
iv. Water.
- b. The rate of transpiration is less when
i. Humidity is high, ii. Atmosphere is dry, iii. Temperature is high,
iv. Windy day.
- c. The process by which water is lost from the margin of a leaf is known as
i. Guttation, ii. Transpiration, iii. Bleeding, iv. Absorption
- d. Transpiration taking place through the minute openings on the surface of old stems
i. Cuticular Transpiration, ii. Lenticular Transpiration,
iii. Stomatal Transpiration, iv. Stoma Transpiration
- e. Sunken stomata is found in
i. *Hibiscus*, ii. *Nerium*, iii. Pine, iv. Bacteria

2. Give reasons for the following: [5]

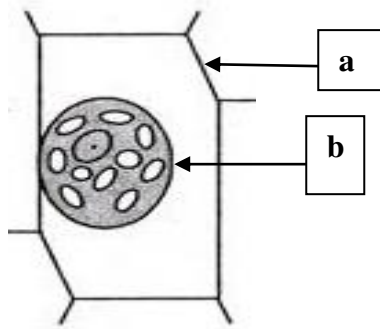
- i. Plasma membrane is known as selectively permeable membrane of a cell.
- ii. Soaked seeds when placed in a fully closed container burst open with great pressure.
- iii. The roots of certain trees have been seen to crack the walls.
- iv. Desert plants have thick cuticle.
- v. Transpiration is greater in a broad leaf than in a narrow leaf.

3. Differentiate between the following according to the criterion given in brackets: [5]
- Evaporation and Transpiration (Definition)
 - Osmosis and Diffusion (Semipermeable membrane)
 - Active Transport and Passive Transport (Definition)
 - Plasmolysis and Deplasmolysis (Definition)
 - Turgidity and Flaccidity (Definition)
4. Draw a neat and well labelled diagram of stomata. Mention how intensity of sunlight and atmospheric pressure affects transpiration. [5]
5. Name the following: [5]
- The non-living cell inclusion inside a cell which contains cell sap.
 - An instrument used to find the rate of transpiration.
 - The solution where the relative concentration of water molecules on either side of the cell membrane is same.
 - The vascular tissue which carries water and minerals from root to different parts of the plant.
 - Absorption of water by the plant cells by surface attraction.
6. Observe the diagram given below and answer the following questions: [5]



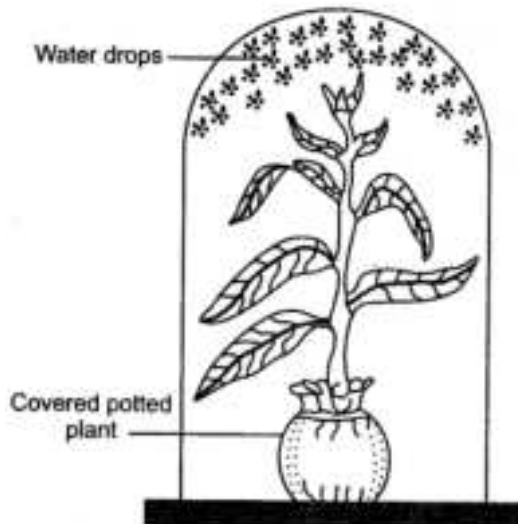
- Identify the instrument.
- Why do we use coloured water in this experiment?
- Mention the utility of water reservoir in this experimental set up.
- Mention two limitations of this instrument.

7. Observe the diagram given below and answer the following questions: [5]



- i. Identify the state of the cell.
- ii. Mention the tonicity of the liquid which is surrounding the cell.
- iii. How will you change the state of the cell?
- iv. Label the parts “a” and “b”.
- v. Mention one example of a semipermeable membrane.

8. Observe the experiment given below and answer the following questions: [5]



- i. Mention the aim of the experiment.
- ii. Why is the pot of the plant covered?
- iii. How will you prove that the droplets of liquid inside the bell jar are water droplets only?
- iv. Suggest a suitable control for this experiment.
- v. Mention one significance of the process shown in this experiment.